

Finding the FrameStack:

Learning What to Remember for Non-Markovian Reinforcement Learning

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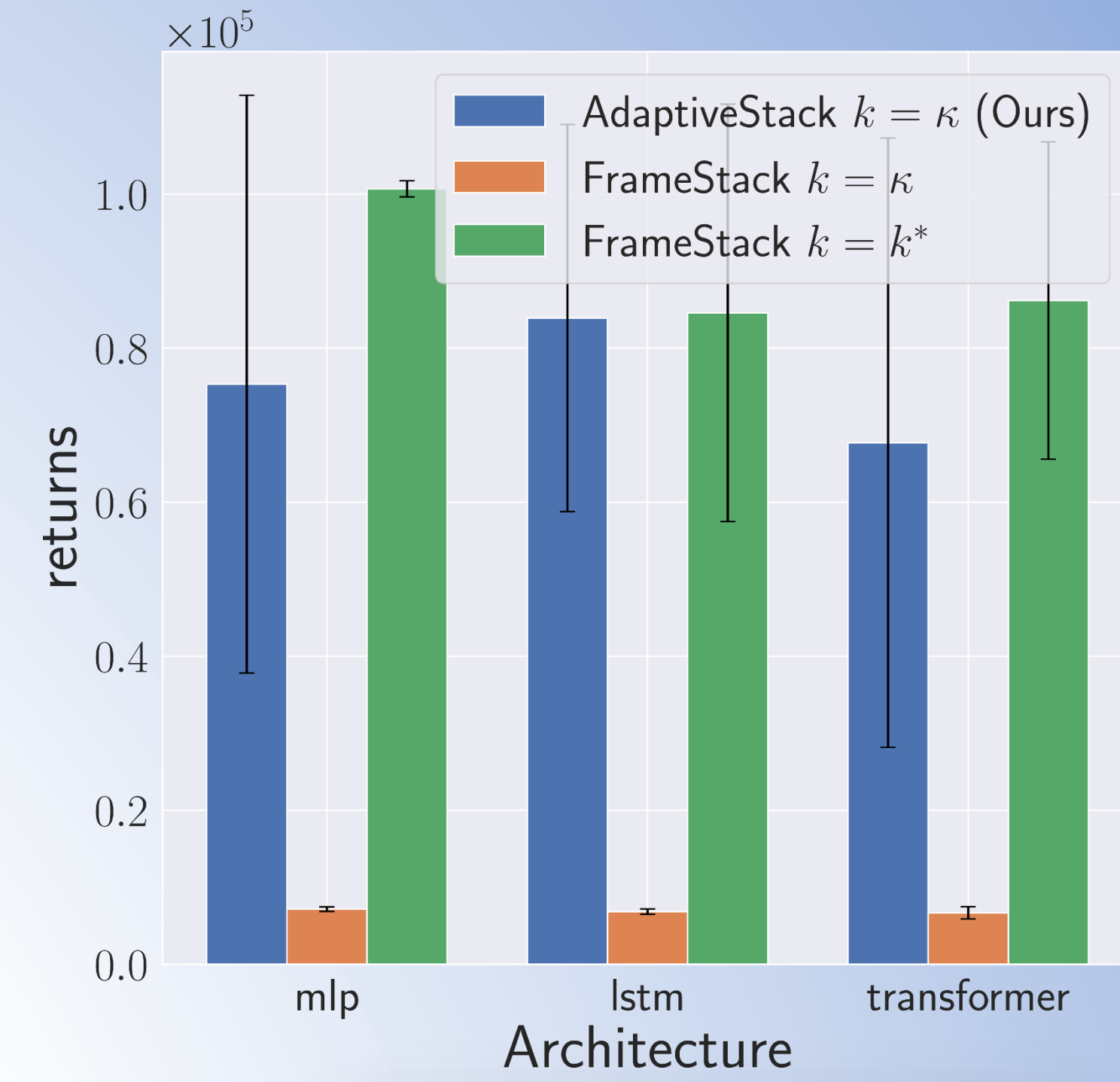
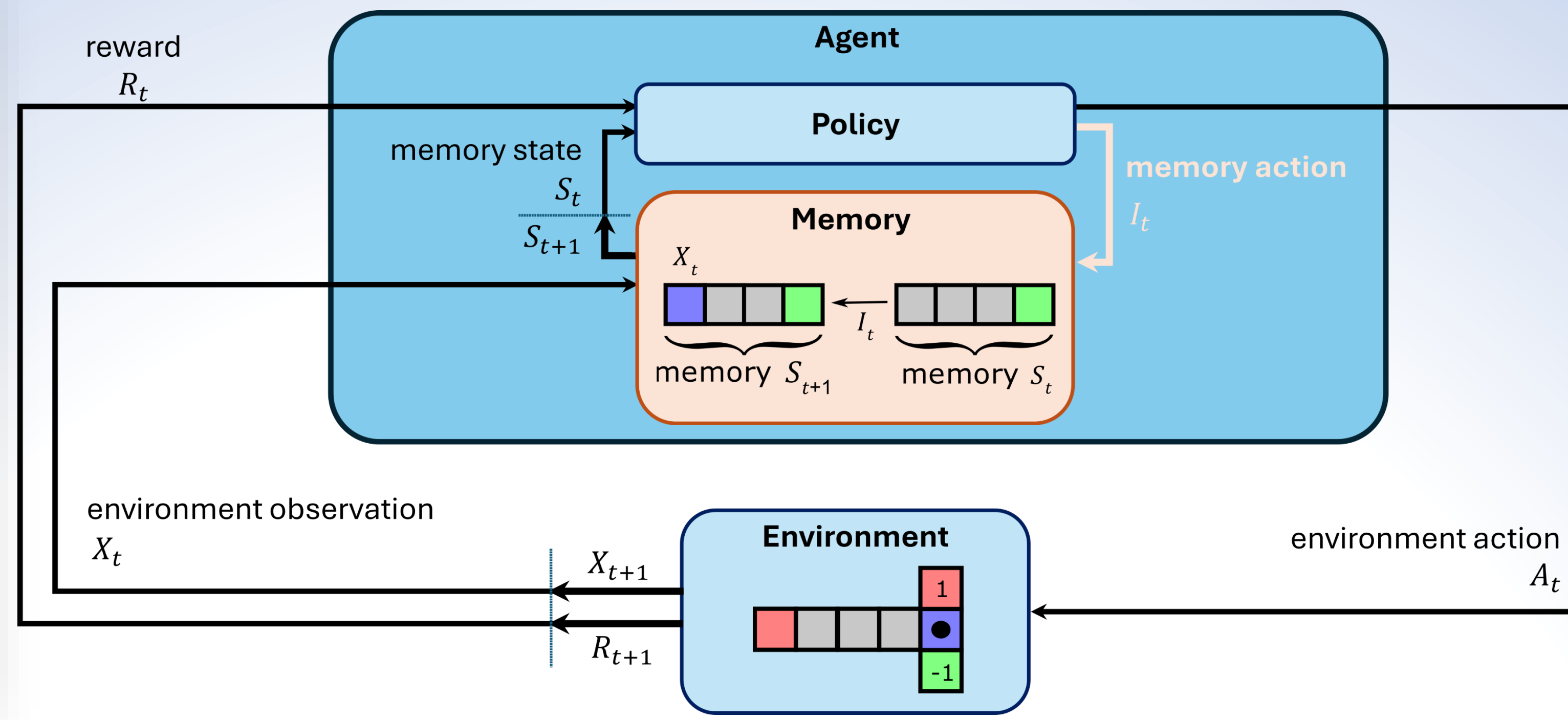
Adaptive Stacking enables RL agents to efficiently handle **long-term dependencies** by learning what to remember, **reducing memory and compute costs** while **preserving optimality**.

Motivation

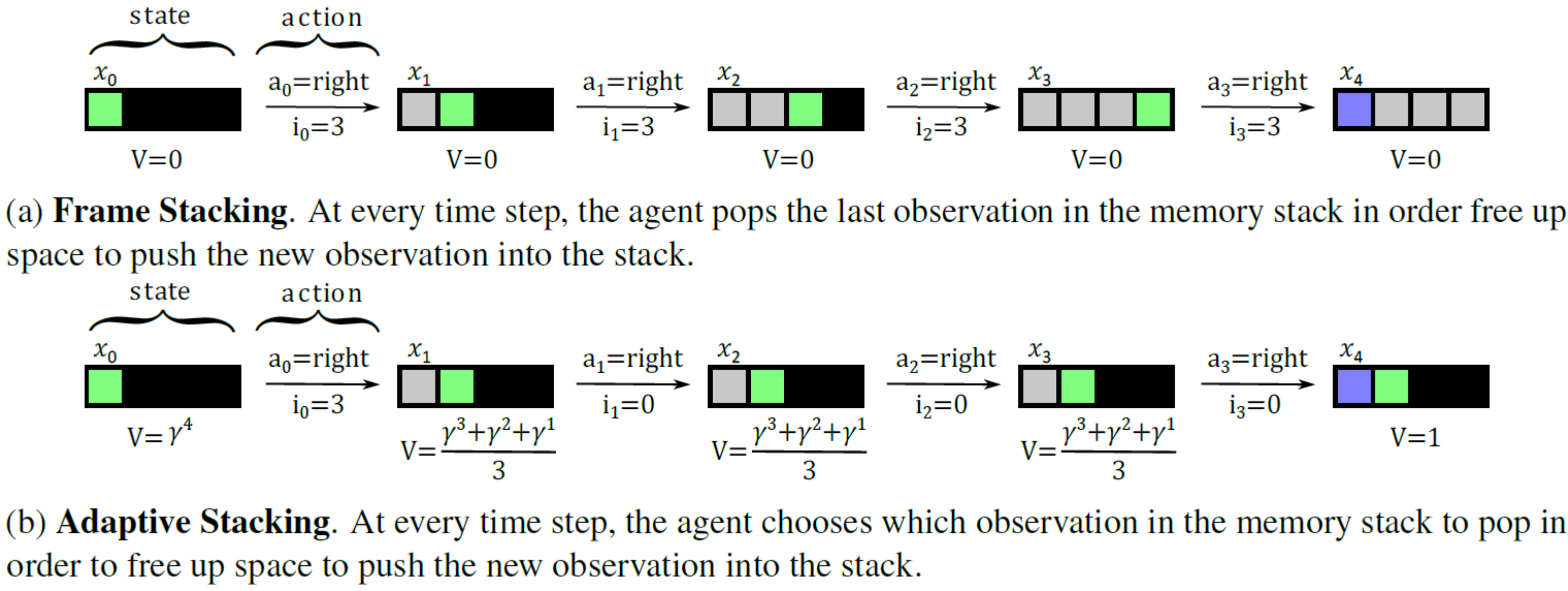
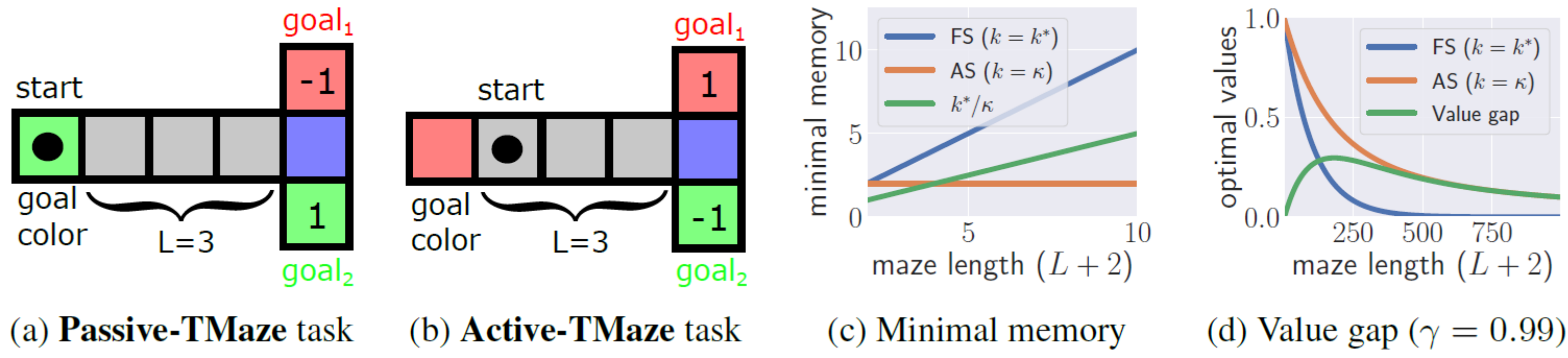
We want a **general-purpose meta-algorithm** for learning to **manage memory** in **partially observable** environments. Unlike standard frame stacking, which blindly retains recent observations, **Adaptive Stacking** allows agents to **learn which observations to remember or discard** via RL.

Exponential increase in observations, which impacts **compute c** and **memory w** ❌

Architecture	Memory Type	$ c _{a \sim \pi_\theta}$	$ c _{TD}$	$ w _{a \sim \pi_\theta}$	$ w _{TD}$
MLP or LSTM	Frame Stack	$\Omega(k^*)$	$\Omega(k^*)$	$\Omega(k^*)$	$\Omega(k^*)$
MLP or LSTM	Adaptive Stack	$\Omega(\kappa)$	$\Omega(\kappa)$	$\Omega(\kappa)$	$\Omega(\kappa)$
Transformer	Frame Stack	$\Omega(k^{*2})$	$\Omega(k^*)$	$\Omega(k^{*2})$	$\Omega(k^*)$
Transformer	Adaptive Stack	$\Omega(\kappa^2)$	$\Omega(\kappa)$	$\Omega(\kappa^2)$	$\Omega(\kappa)$

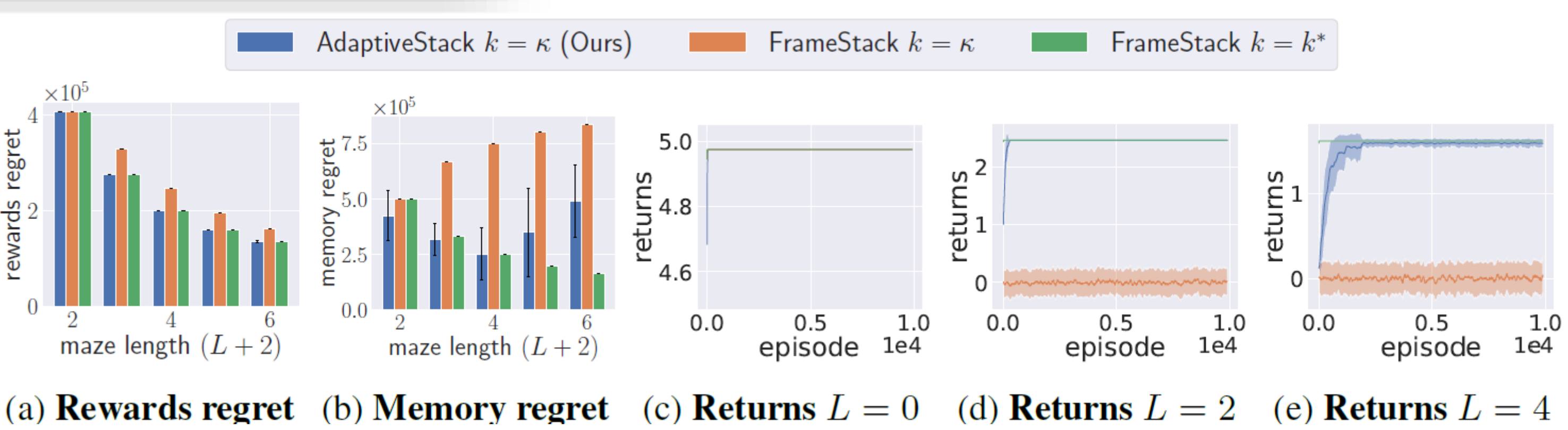


RL with Internal Memory Decisions

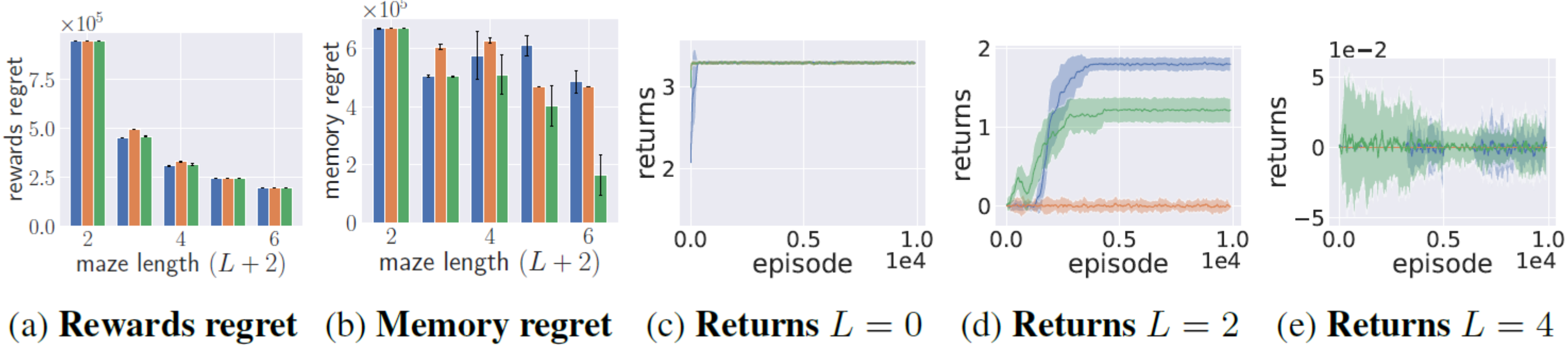


$$V_2^{\pi^*}(\text{state}) = \frac{1}{3} V_2^{\pi^*}(\text{state}) + \frac{1}{3} V_2^{\pi^*}(\text{state}) + \frac{1}{3} V_2^{\pi^*}(\text{state}) = \frac{1}{3} (\gamma^3 + \gamma^2 + \gamma)$$

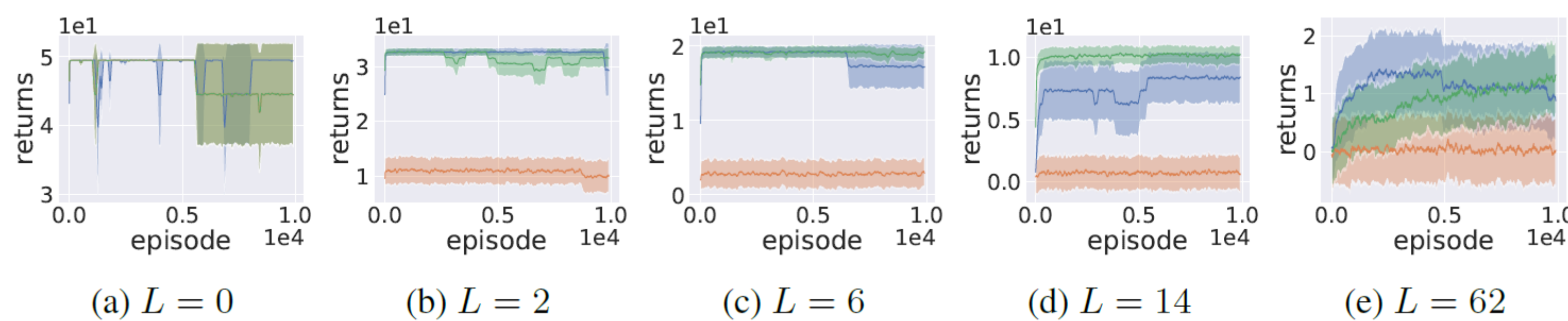
Training



Continual Passive-TMaze with Q-learning



Continual Active-TMaze with Q-learning



Episodic Passive-TMaze with PPO using an MLP

Adaptive Stacking

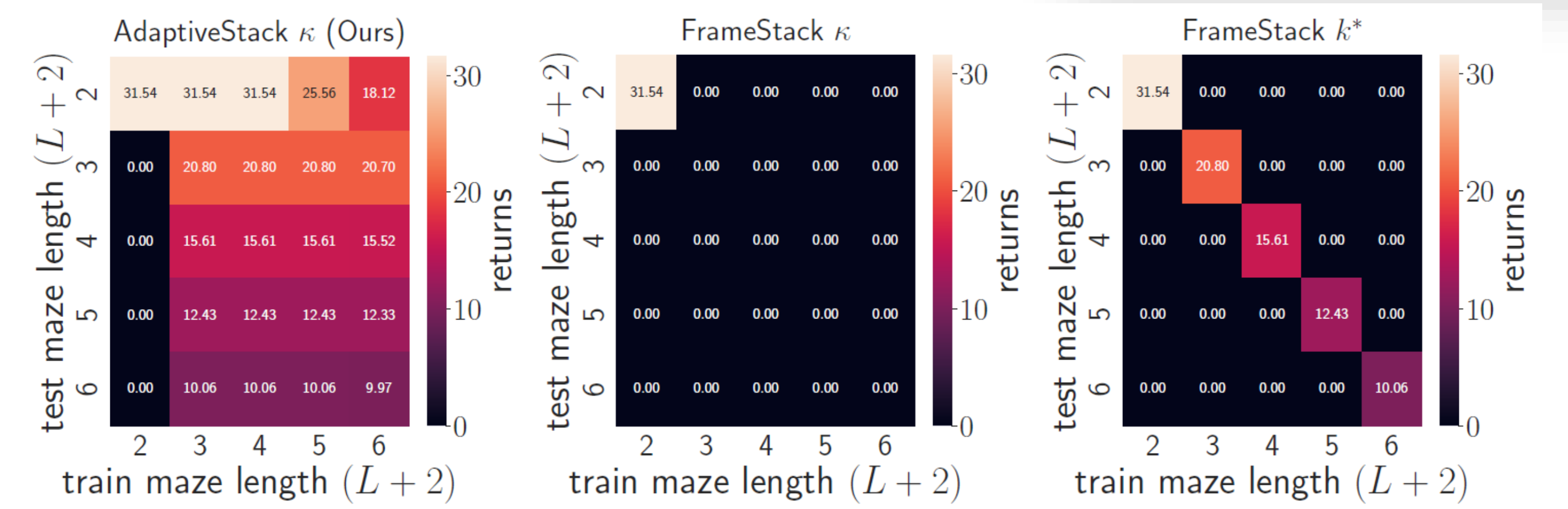
Get initial observation $x_0 \in \mathcal{X}$
Initialise observation stack $s_0 \leftarrow [x_0]_k$
foreach timestep $t = 0, 1, \dots, T$ **while** episode is not done **do**
 $\langle a_t, i_t \rangle \leftarrow \begin{cases} \arg \max_{(a,i)} Q(s_t, \langle a, i \rangle) & \text{w.p. } 1 - \epsilon \\ \text{a random action} & \text{w.p. } \epsilon \end{cases}$
 Execute a_t , get reward r_{t+1} and next observation x_{t+1}
 Remove observation from stack $s_{t+1} \leftarrow \text{pop}(s_t, i_t)$
 Push observation into stack $s_{t+1} \leftarrow \text{push}(s_{t+1}, x_{t+1})$

Remark 1 Uncertainty in history may harm value expectations, $|V^*(x_{t:t-k^*}) - V_k^{\pi^*}(s_t)| > 0$, but it does not necessarily harm policy optimality as long as the uncertain differences are irrelevant for optimal decision making: $V^*(x_{t:t-k^*}) = V_k^{\pi^*}(x_{t:t-k^*})$.

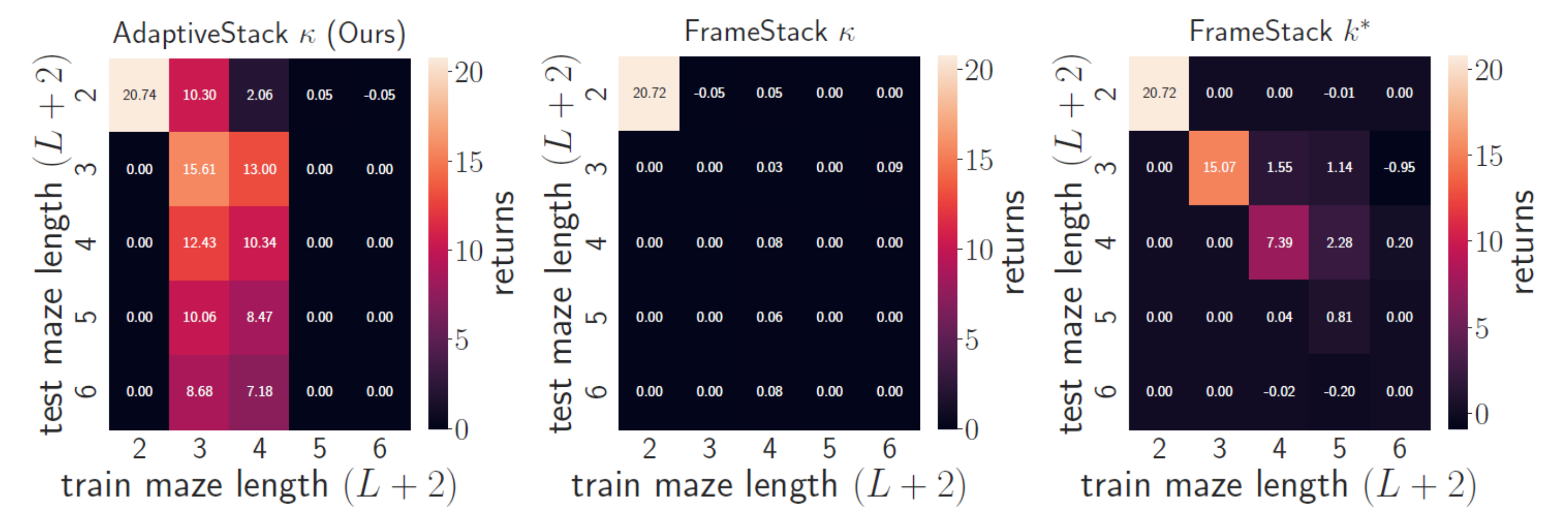
Definition 1 Define κ to be the smallest memory length such that there exists a policy π_κ^* satisfying $V_k^{\pi_\kappa^*}(x_{t:t-k^*}) = V^*(x_{t:t-k^*})$ for all t .

Theorem 1 Let $\mathbb{A}$ be an RL algorithm that converges under Frame Stacking with $k \geq k^*$. If $\mathbb{A}$ uses unbiased value estimates to learn optimal policies, then it also converges under Adaptive Stacking with $k \geq \kappa$ observations, assuming the policy class is sufficiently expressive.

Evaluation



Continual Active-TMaze with Q-learning



Episodic Passive-TMaze with PPO using an MLP

